

Nexus RAG

① Architecture layers ✓✓

② Team pod ✓✓

③ Cost. ✓✓

Architecture :-

① Ingestion layer :-

✓① HR ✓

✓② Fin ✓

✓③ legal ✓

✓④ mom ✓

✓⑤ Research Paper.

✓⑥ Code.

✓⑦ Project / Product PRD, LLD, HLD

1000 Emp , 5 year , 1M docs. $\approx$ 25mb-30mb

- ✓ ① Sharepoint
- ✓ ② Google Drive
- ✓ ③ One drive
- ✓ ④ S3 bucket
- ✓ ⑤ Azure Blob
- ✓ ⑥ SQL DB
- ✓ ⑦ No-SQL DB

Team - work

- ① write a connector
- ② scheduler
- ③ watcher.

PDF, doc, pptx, txt, ppt, audio,

① image

② mult-language.

③ meta-data management

level-2

- ① Process diff files.
- ② image
- ③ multi lang.
- ④ meta-info
- ⑤ Audio

level-3

② Indexing of data :

① Chroma DB / Pine Cone
//

② Deduplication / versioning.

③ chunking / overlapping.

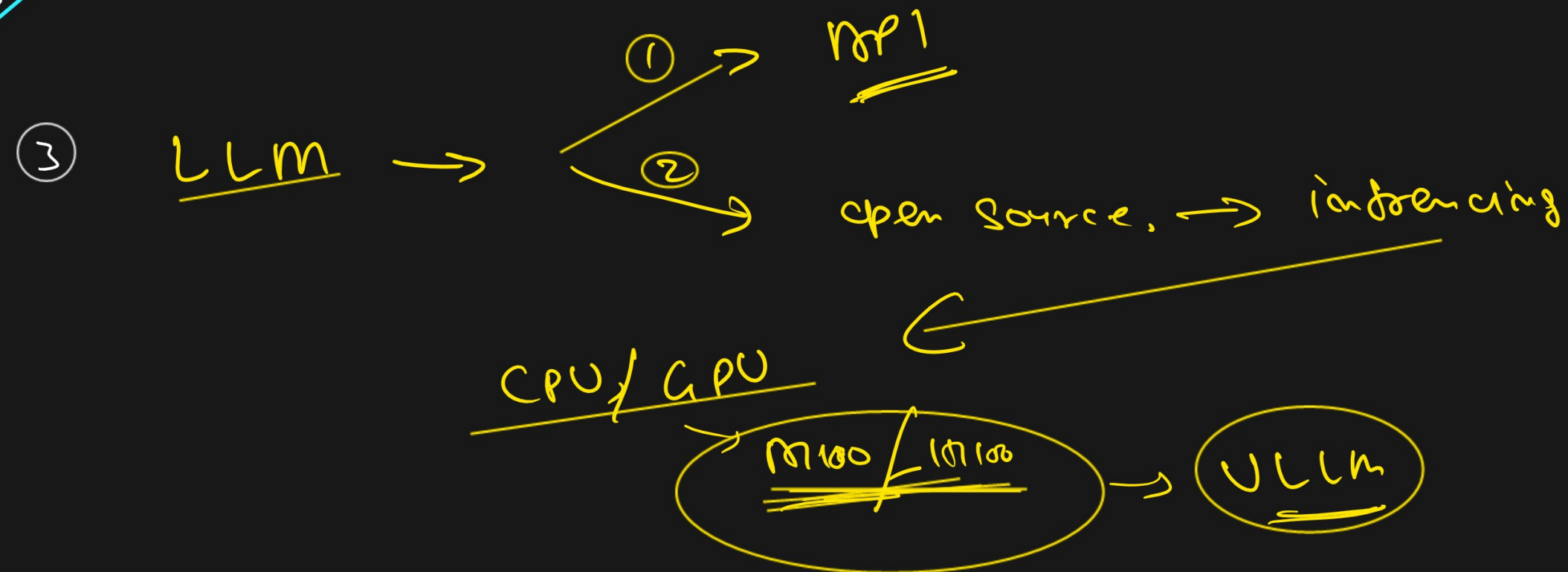
④ Embed models.
// //  API ←
local model. / open source ←

⑤ vector size → $\frac{1536}{416}$

③ Retrieval & RAG Layer

① Hybrid (C_{og}^{row} + B_m²⁵) → 30

② Reformer



④ Context optimization (lang chain / LLMs indices)

④ Application layer :-

① web App

② mobile App

③ slack / teams / whatsapp / telegram

④ Admin Dashboard. → role based Access

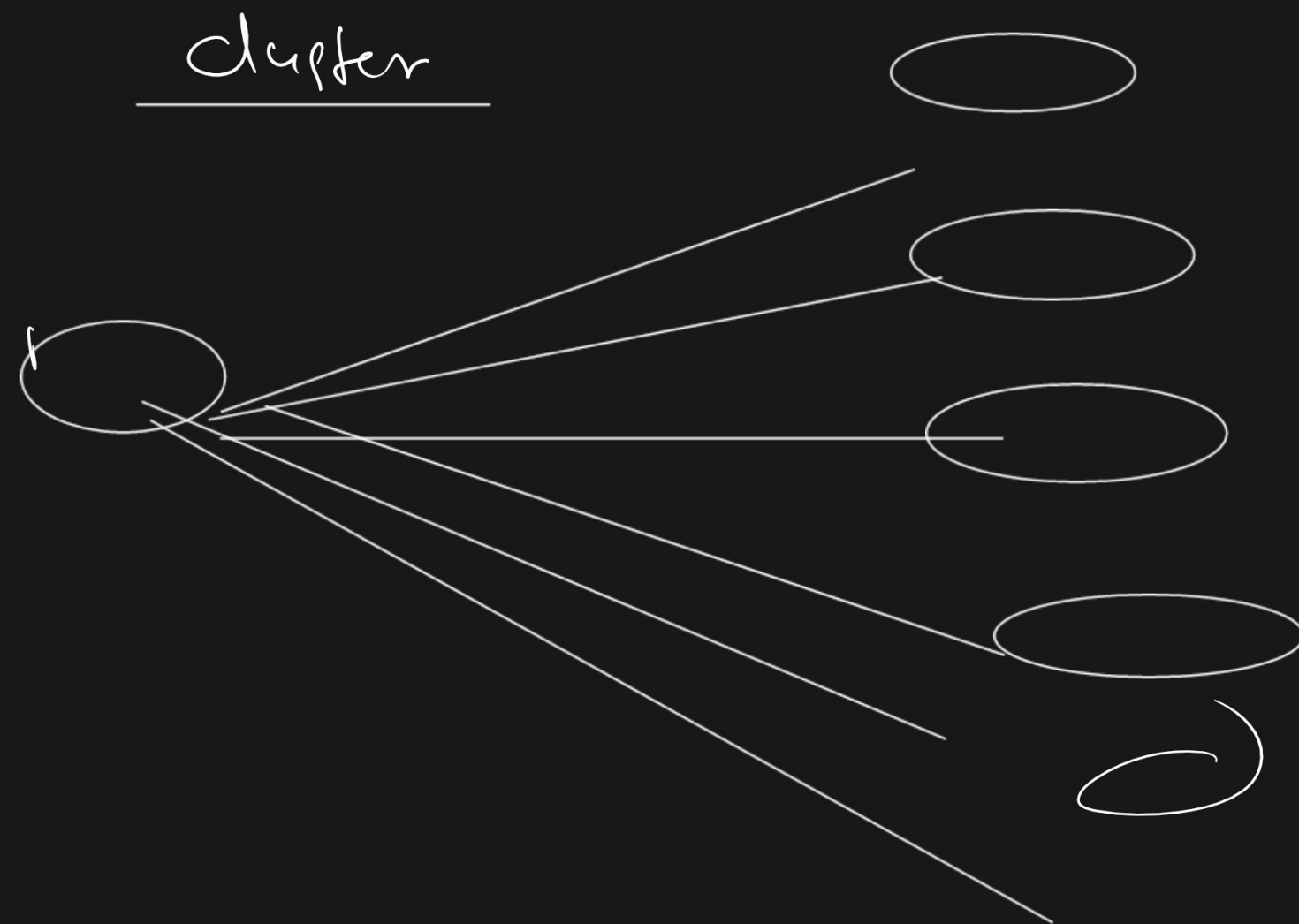
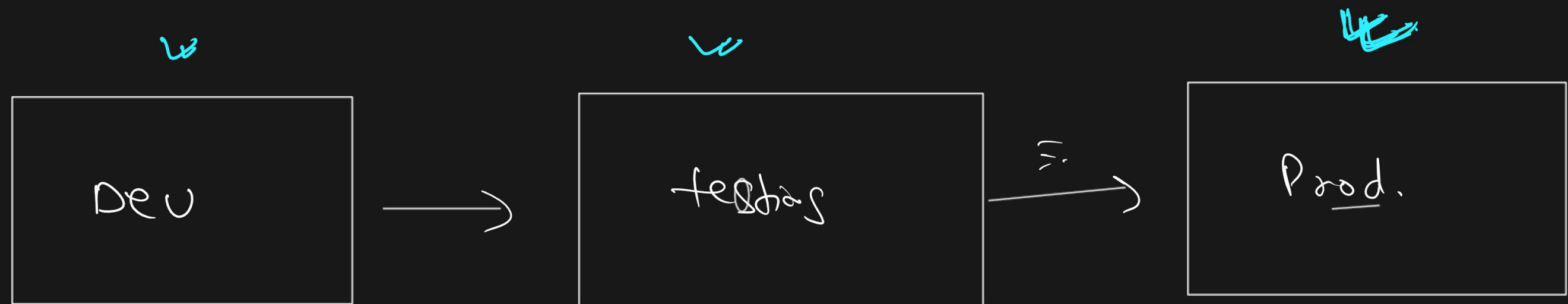
⑤ Security layer :-

① SSL

② E-2-1 Encryption.

③ Audit log & monitoring.

⑥ Deployment layer



<u>Cost bucket</u>	S	M	L
user	100	1000	10000
data	50k docs	500k docs	~ 5m docs
OPS	2-3	15-20	100-200
node	3	6	15
capacity	avg. large (2ucpu, 8GB)	avg	avg
EKS	\$201	\$328	\$711
UDS → same core	\$500	\$500	\$1256
S3 →	\$5 (200GB)	\$47 (2TB)	\$471 (20TB)

LLM - api
GPT-5 min

\$ 112

\$ 1125

\$ 11250

Behavior
(coherence)

\$ 30

\$ 300

\$ 3000

\$ 1258 / mo

\$ 3601 / mo

\$ 24481 / mo

① ETL & Data Pipeline

Streaming → Kafka / AWS Kinesis

Data Processing → PySpark / DataBricks

Scheduler → Airflow

② Indexing & Storage

Vector DB

- ① FAISS
- ② Chroma DB
- ③ Pinecone
- ④ Weaviate
- ⑤ Qdrant

Tradition Search

- ① Elastic Search
- ② open Search.
- ③ veps

Storage

- ① S3 / Azure Blob/CS

③ Embedding & Re-ranking

- ① $E_{u, v}$ DPF $\rightarrow$
 - ② open m
 - ③ cohere embedding
 - ④ Hugging face sentence transformers
- $\Rightarrow$ Embedding

Re-ranking :-

- ① BGE - re-ranker
- ② Cohere re-ranker
- ③ Hugging face

④ RAG & LLM

Framework

- ① langchain
- ② llama index
- ③ langgraph
- ④ crew ai

LLM

EUROPI → (openai, Deepseek, mist, Gemini)

openai GPT →

Gemini —

meta LLaMA —

Anthropic Claude —

⑤ Application layer

Backend → FastAPI / Flask / Django Rest

Frontend → React / Next.js

↳ Photo typing → Streamlit / Gradio

Deployment

① EC2

② API Gateway / load balancing → Amazon, AWS, API, Gateway

③ CI/CD → GitHub Actions / GitLab CI / Jenkins.

monitoring

- ① metrics & alerts $\rightarrow$ Prometheus + Grafana
- ② logging $\rightarrow$ ELK stack / open telemetry
- ③ Tracing $\rightarrow$ Jaeger / Zipkin.

Team structure

① Core team

① Product manager

② Solution Architect

③ Tech lead → multi stack wide

Pods

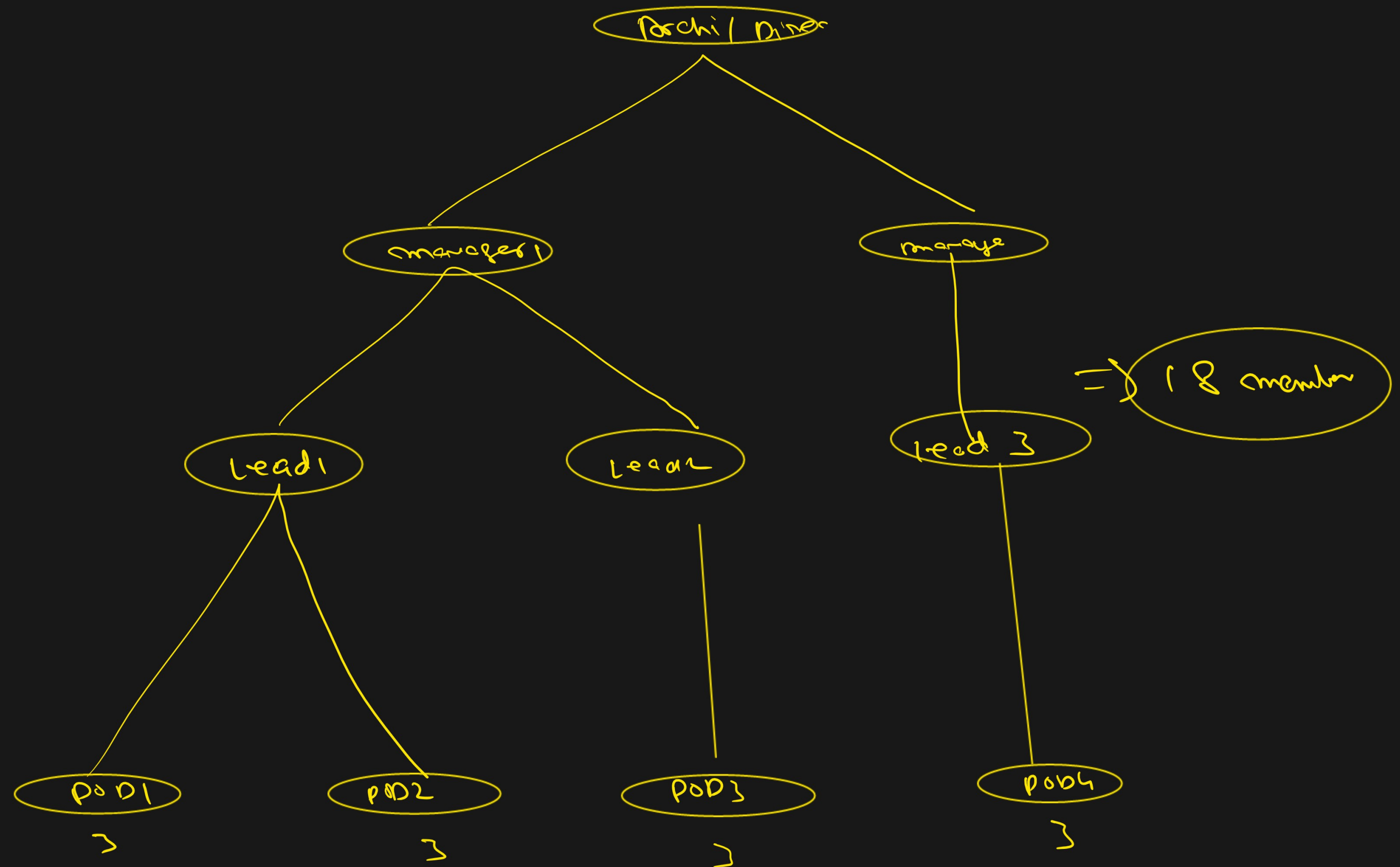
① ingestion pod ✓

② Retrieval pod ✓

③ ~~DB~~ 2 ingestion pod ✓

④ Platform & sec pod ✓

⑤ QA 2 Analytics POD ✓



Estimate timeline → 6 months

Ph-1, 2, 3, 4, 5, 6
↓ ↓ ↓ ↓ ↓ ↓
4 weeks